

Impressioning Attack on an Ancient Egyptian Lock

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Abstract

The first key-operated lock in history is the ancient Egyptian pin-tumbler lock. It was a large wooden lock, commonly made from the wood of a fig tree. The design of this lock continues to have a tremendous influence on modern locks. There seems to be no record, however, of any successful non-destructive attack on a lock of this type. We feel that an investigation into plausible methods of defeating such a lock is long overdue.

TL;DR – The lock can be defeated in minutes using sticks and mud.

The earliest known use of pin-tumbler locks was in Egypt, around 2000 BC. These locks were large, because large pieces of wood are harder to break than small ones. There's no record of lockpicking in that era, neither by criminals nor by the carpenters who served as locksmiths until the redsmiths and blacksmiths took over that role a couple of millennia later. But we are unconvinced that this means it didn't happen.

We set out to investigate how these locks work and how carpenters, crooks, and spies might have defeated them when necessary. In the absence of definitive evidence, we'll settle for proving vulnerability to a historically plausible method.

The Lock

The subject of this experiment is a 3D-printed replica of an ancient Egyptian lock. Having studied historical records for a detailed understanding of the particulars of such a lock, we selected a geometric model by [Str8_3D] of South Africa as an exemplary specimen. We printed the parts at 200% scale and assembled them into two such locks for experimentation. One of these is pictured in Figure 1.

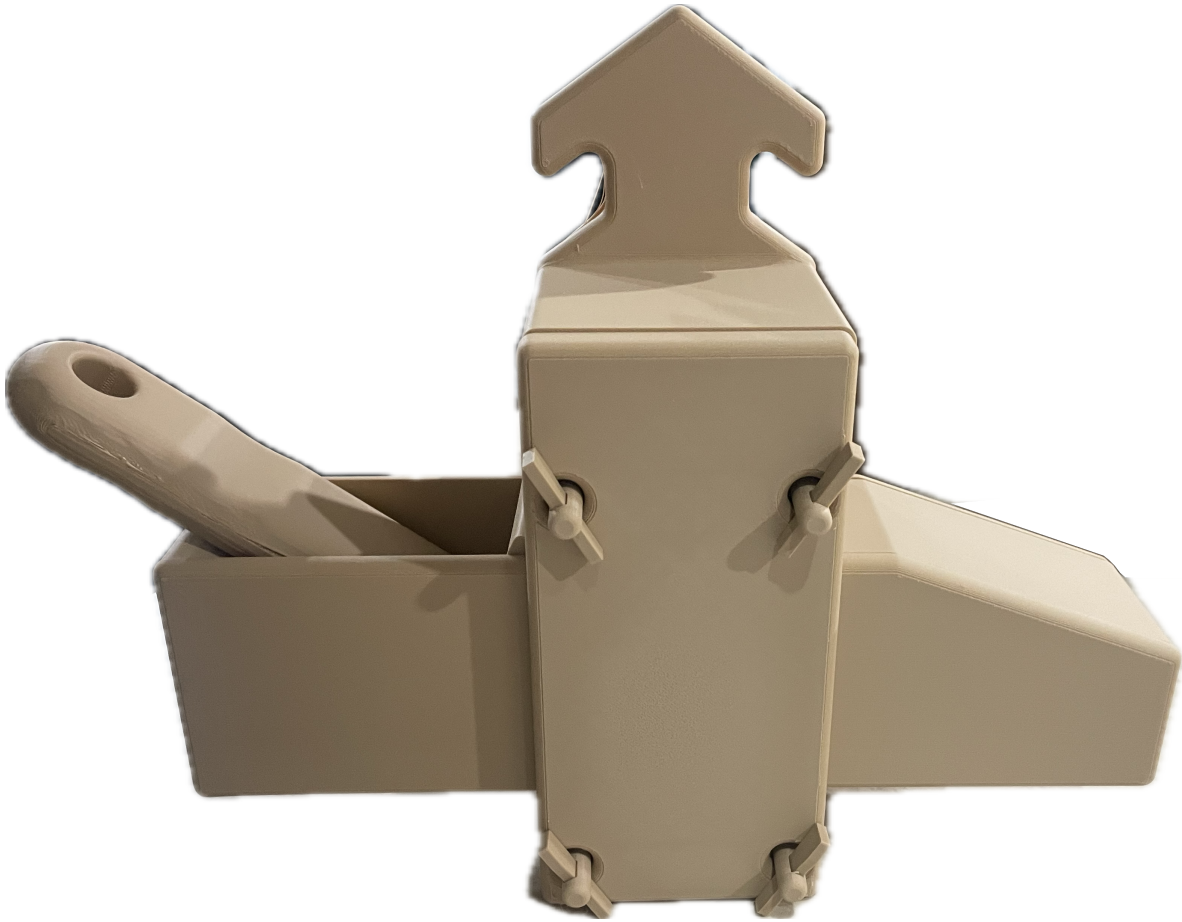


Figure 1: Replica of an Ancient Egyptian Lock

Each lock has five pin chambers. (An ordinary number, but this was known to vary from three to twelve.) We observe that these pin chambers and the pins that occupy them differ from their modern counterparts in the following ways:

1. Only the top pins (the tumblers) are integral parts of the lock. Each chamber's bottom pin (the bit) is part of the key instead.
2. All bits of a key have the same height. That height is not a secret but may vary somewhat between locks, especially between locks from different makers.
3. Rather than varying the height from bit to bit, each key's secret bitting is encoded through the positioning of its bits on the horizontal plane.
4. Each tumbler is driven downward only by gravity, with no spring assistance.
5. The pins and pin chambers are enormous! – 5mm diameter as created for this experiment, perhaps larger historically.
6. Each pin chamber has two segments, one in the lock case and the other in the bolt.
7. Below the pins' lower chambers is a cutout within the bolt, to accommodate insertion of the business end of the key handle.
8. Each tumbler has an enlarged top, to keep it from falling entirely out of its chamber.
9. When it was necessary for the lock to be operable from both sides of the door, the lock was traditionally mounted on the inside and the door was given an armhole to reach through.

Figure 2 demonstrates this lock in action.

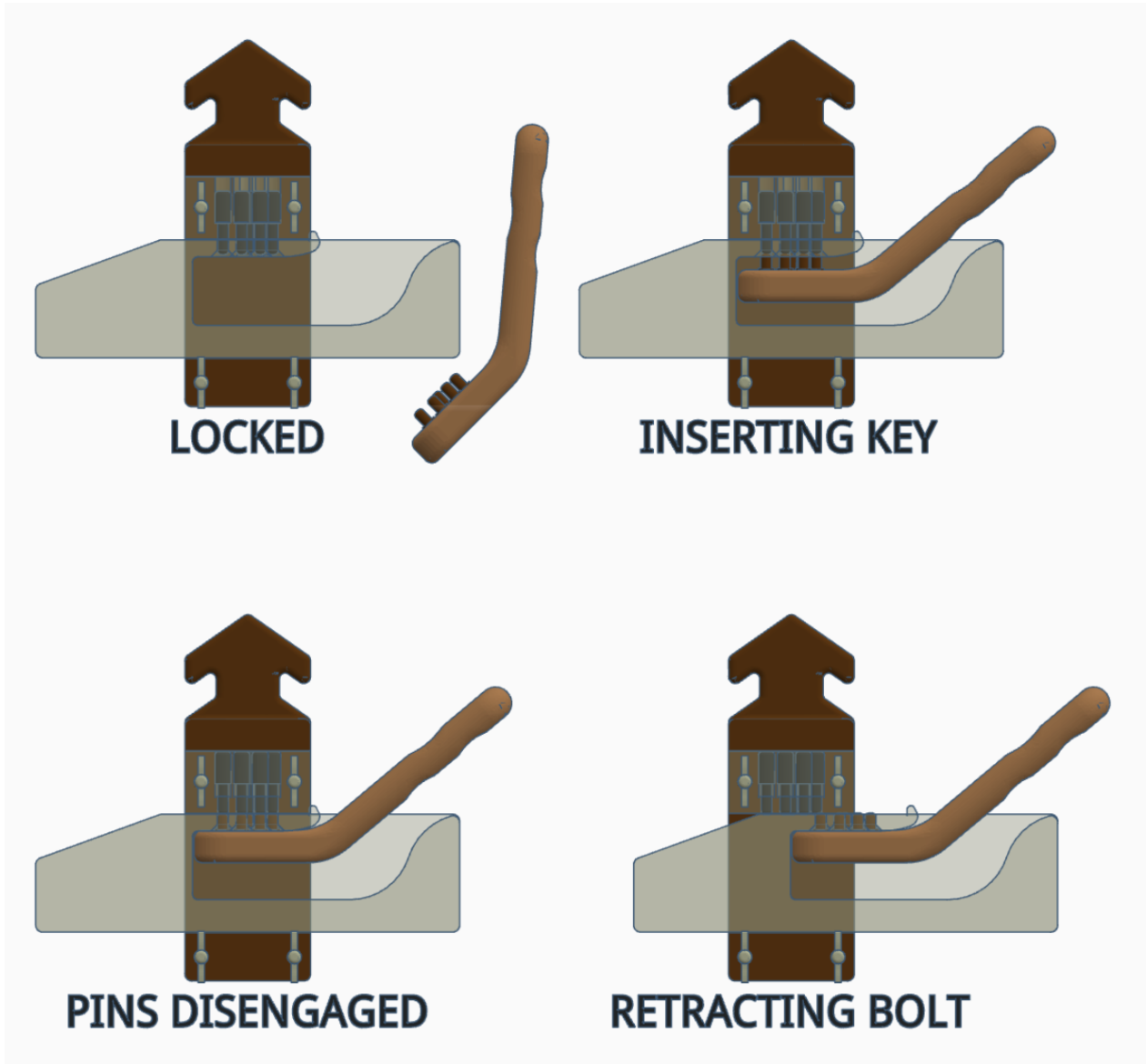


Figure 2: Operation of the Lock

The Attacks

We devised in our minds various impressioning attacks against this lock. Impressioning is an interactive skill of causing a lock to impress clues about itself onto a partial key and modifying the key according to those clues. To our knowledge, impressioning has never before been performed on any lock of the Egyptian style. We believe, however, that it is in fact better suited to such a lock than to any other style.

Here we present three historically feasible versions of an impressioning attack against this lock:

1. one for a carpenter – the neatest and most durable but rather slow
2. one for a burglar – crude and fragile, but tends to get the job done fairly quickly
3. one for a spy – a refined approach that's reliable and the fastest of all

We describe here all three, with a full demonstration of the spy's attack because we see it as the most advantageous variant.

The Carpenter's Attack

The limited history suggests that perhaps no ancient carpenter ever devised a way to defeat these wooden locks non-destructively, but we find this implausible. After all, people must have been reasonably skilled at losing their keys back then, and it would behoove a skilled lock builder in any age to find a solution.

These locks are not so complex as to make it infeasible to identify the pattern of pins required in a key for any given lock. One such method that we devised, suitable for an ancient carpenter, is as follows:

1. Take from your stock a blank key: a wooden handle without bits.
2. Coat the top surface of the key with a thin layer of clay or mud.
3. Insert the key into the bolt's cutout.
4. Press the key up, briefly, against the tumblers.
5. Remove the key.
6. Note the locations of the mounds of clay.
 - They correspond to the tumbler positions, where the pin chambers and freely moving tumblers allowed the clay to remain uncompacted by the bolt's ceiling.
7. Use your bow drill to create holes in the key at each tumbler position.
8. Glue a wooden pin into each of the key's holes to serve as a bit.
 - Err on the side of bits that may be slightly too long.
 - Or perhaps, if you made the lock yourself or are familiar with the maker, you'll know exactly how long to make them.
9. Use the completed wooden key to operate the lock.
 - If the bits are slightly too long, then they'll over-lift the tumblers and the bits themselves will block retraction of the bolt.
 - In that case, you'll find that you have to lower the key slightly from its full upward position to retract the bolt.
10. With the bolt retracted and its bottom pin chambers visible, fine-tune the length of each bit if necessary for a perfectly smooth-operating key.

We estimate that this process would take a skilled carpenter about twenty minutes to complete with the tools of that era. It offers the carpenter advantages over destructive methods, producing a key in the process and avoiding damage to the lock or door that would require his labors to repair. A carpentry guild in possession of the knowledge to perform it may have safeguarded this secret, in the interest of maintaining the usefulness and reputation of the locks.

Security professionals of the past often considered this sort of "security by obscurity" to be a suitable way of protecting society. This very well may be the only reason why ancient carpenters didn't leave documentation of attacks such as this. We consider such safeguarding of information to be a sketchy approach. It misleads consumers about what protection the lock provides, and it depends on the fragile assumption that burglars and spies will never catch on.

The Burglar's Attack

Lack of historical evidence also suggests that perhaps no ancient *burglar* ever devised a method of non-destructive entry either. That's plausible, as a burglar doesn't tend to concern himself much with avoiding damage. But a clever one *would* favor a quick and quiet entry over a slow and noisy axe. So we devised a plausible speed-optimized variant of the carpenter's attack.

1. Find a bent stick, approximating the curve of a key handle.
2. Place a large wad of clay or thick mud on the part of the stick which will interact with the tumblers.

3. Insert the stick into the bolt's cutout as you would insert a key.
4. Press the stick up, briefly, against the tumblers.
5. Remove the stick.
6. Note the locations of the mounds of clay.
 - They correspond to the tumbler positions, where the pin chambers and freely moving tumblers allowed the clay to remain uncompacted by the bolt's ceiling.
7. Press a small stick fragment into each mound of clay to serve as a bit.
 - Since you may be unable to accurately guess the proper bit length, err on the side of bits that may be too long.
8. Use the improvised key that you've created to operate the lock.
 - If the bits are slightly too long, then they'll over-lift the tumblers and the bits themselves will block retraction of the bolt.
 - In that case, you'll find that you have to lower the key slightly from its full upward position to retract the bolt.

A burglar in possession of the knowledge to perform this attack would likely have kept it to himself, in the interest of not getting found out.

First Execution of the Burglar's Attack

The burglar's variant of our impressing attack was first successfully executed against an Egyptian-style lock by Liam and Isaac in Meridian, Idaho, at 7:54pm MST on January 28, 2026, using pieces of a broken pencil as the bits. (See Figure 3.) It took eight minutes. To our knowledge, this is the first documented non-destructive opening of any lock of ancient Egyptian design. The attack was subsequently reproduced by others, who found popsicle sticks easier to work with.



Figure 3: The Burglar's Attack

The Spy's Attack

Spies are the most secretive of all, so it comes as no surprise that the spies of ancient Egypt didn't leave a record of their exploits. But a well-funded agency of the Pharaoh's spies could plausibly have developed the means to surreptitiously defeat these locks and would certainly have had an interest in doing so.

1. Start with a specialized impressionist key blank and bits, prepared in advance.
 - We 3D-printed our specialized blank, because we're handier with 3D printers than we are with woodworking.
 - The specialized blank should contain a bed of clay in the biting area, extending from deep inside it to slightly above its rim.
 - The bits should be smaller in diameter than traditional key bits, as this makes it easier to produce a key that operates smoothly.
2. Insert the key handle into the bolt's cutout as you would insert a key.
3. Press the key handle up, briefly, against the tumblers.
4. Remove the key handle.
5. Note the locations of the mounds of clay.
 - They correspond to the tumbler positions, where the pin chambers and freely moving tumblers allowed the clay to remain uncompacted by the bolt's ceiling.
6. Press a prepared bit into each mound of clay.
 - If you're familiar with the lock's maker, then it may be possible to make the bits of exactly the right length. Otherwise, err on the side of slightly overlength bits.
7. Use the key that you've created to operate the lock.
 - If the bits are slightly too long, then you'll find that you have to lower the key slightly from its full upward position to retract the bolt.
8. With the bolt retracted and its bottom pin chambers visible, shift each bit by hand into the center of its chamber.
9. Note whether any fine-tuning of bit lengths is necessary for a perfectly smooth-operating key, and lift or trim the bits accordingly.
10. Remove the key and let it dry for future re-use.

This process takes us about two minutes to complete, and it results in a durable key that can be reused instantly in the future. A spy agency in possession of the knowledge to perform it would certainly have kept it a secret, preferring to keep their capabilities unknown to the enemy.

Demonstration of the Spy's Attack

The spy's variant of our impressioning attack was first successfully executed against an Egyptian-style lock by Jeremy Reeder in Meridian, Idaho, at 10:24pm MST on January 28, 2026, using segments of a lollipop stick as the bits. (See Figures 4 thru 6.) It took three minutes.



Figure 4: A Spy's Key, With a Clay-Filled Bed



Figure 5: Impressions & Bits on a Spy's Key



Figure 6: Bolt Retracted by a Spy's Key

Conclusion

Impressioning is a very effective attack method against ancient Egyptian pin-tumbler locks. It offers advantages for carpenters, burglars, and spies alike. For spies we believe it is the most practical attack method. We enjoy imagining how the Pharaoh's spies may have employed it. We've demonstrated the feasibility of this, but it remains a mystery whether this was actually done. In the absence of historical records to prove that it was, we declare ourselves as history's first known impressioners of this type of lock.

The Victors

- Jeremy Reeder – Certified Professional Locksmith
- Liam Weeks – Apprentice Locksmith
- Isaac (who prefers to remain un-surnamed) – Apprentice Locksmith

References

[Str8_3D] *Model of Ancient Egyptian Lock*. <https://www.printables.com/model/306746>